NATIONAL UNIVERSITY OF SINGAPORE

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MA1522 Linear Algebra for Computing **Graded Quiz 5**

1. Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$.

(1) If $\boldsymbol{u}$ is orthogonal to $\boldsymbol{v}$, then $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2$.

(2) If $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2$, then $\boldsymbol{u}$ is orthogonal to $\boldsymbol{v}$.

Solution. Both (1) and (2) are true.

$$\|\boldsymbol{u}+\boldsymbol{v}\|^2 = (\boldsymbol{u}+\boldsymbol{v})\cdot(\boldsymbol{u}+\boldsymbol{v}) = \boldsymbol{u}\cdot\boldsymbol{u} + 2\boldsymbol{u}\cdot\boldsymbol{v} + \boldsymbol{v}\cdot\boldsymbol{v} = \|\boldsymbol{u}\|^2 + 2\boldsymbol{u}\cdot\boldsymbol{v} + \|\boldsymbol{v}\|^2.$$

Thus $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2 \Leftrightarrow \boldsymbol{u}\cdot\boldsymbol{v} = 0 \Leftrightarrow \boldsymbol{u}$ is orthogonal to $\boldsymbol{v}$.

2. Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$.

(1) $\|\boldsymbol{u}+\boldsymbol{v}\| \le \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$.

(2) $\|\boldsymbol{u}+\boldsymbol{v}\| \ge |\|\boldsymbol{u}\| - \|\boldsymbol{v}\||$.

Solution. (1) True.

$$\begin{aligned}\|\boldsymbol{x}+\boldsymbol{y}\|^2 &= (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y}) = \boldsymbol{x}\cdot\boldsymbol{x} + 2\boldsymbol{x}\cdot\boldsymbol{y} + \boldsymbol{y}\cdot\boldsymbol{y} \\ &= \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\|\cos\theta + \|\boldsymbol{y}\|^2 \le \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 = (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2,\end{aligned}$$

where θ is the angle between $\boldsymbol{x}$ and $\boldsymbol{y}$.

Since both $\|\boldsymbol{x}+\boldsymbol{y}\|$ and $\|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ are non-negative, the inequality is preserved when taking square-roots, so that $\|\boldsymbol{x}+\boldsymbol{y}\| \le \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

(2) True. Let $\boldsymbol{w} = \boldsymbol{u}+\boldsymbol{v}$. Then

$$\|\boldsymbol{u}\| = \|\boldsymbol{w} + (-\boldsymbol{v})\| \le \|\boldsymbol{w}\| + \|-\boldsymbol{v}\| = \|\boldsymbol{w}\| + \|\boldsymbol{v}\|;$$

$$\|\boldsymbol{v}\| = \|\boldsymbol{w} + (-\boldsymbol{u})\| \le \|\boldsymbol{w}\| + \|-\boldsymbol{u}\| = \|\boldsymbol{w}\| + \|\boldsymbol{u}\|.$$

Thus, $\|\boldsymbol{w}\| \ge \|u\| - \|v\|$, and $\|\boldsymbol{w}\| \ge \|v\| - \|u\|$.

3. Let S be a subset of $\mathbb{R}^n$.

(1) If S is orthogonal, then S is linearly independent.

(2) If S is orthonormal, then S is linearly independent.

Solution. (1) False. $\{\mathbf{0}\}$ is orthogonal, but not linearly independent.

(2) True; see Pg 24 of Chapter 5.

4. Let S be a basis for $\mathbb{R}^n$, and let $\boldsymbol{A}$ be an $n \times n$ matrix whose rows are precisely the elements of S.

(1) If S is orthogonal, then $\boldsymbol{A}$ is orthogonal.

(2) If S is orthonormal, then $\boldsymbol{A}$ is orthogonal.

Solution. (1) False. $\{(2,0),(0,1)\}$ is an orthogonal subset of $\mathbb{R}^2$, but $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ is not orthogonal.

(2) True; see Pg 34 of Chapter 5.

5. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n.

(1) If both $\boldsymbol{A}$ and are orthogonal, then $\boldsymbol{AB}$ is orthogonal.

(2) If $\boldsymbol{AB}$ is orthogonal, then both $\boldsymbol{A}$ and $\boldsymbol{B}$ are orthogonal.

Solution. (1) True. $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1} = \boldsymbol{B}^T\boldsymbol{A}^T = (\boldsymbol{AB})^T$.

(2) False. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. Then $\boldsymbol{A}$ and $\boldsymbol{B}$ are not orthogonal, but $\boldsymbol{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is orthogonal.

6. Let $\boldsymbol{A}$ be an orthogonal matrix of order n. How many possible values of $\det(\boldsymbol{A})$ are there?

Solution. Answer: 2. $1 = \det(\boldsymbol{I}_n) = \det(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}) = \det(\boldsymbol{A}^{\mathrm{T}})\det(\boldsymbol{A}) = (\det(\boldsymbol{A}))^2$. Thus $\det(\boldsymbol{A}) = \pm 1$.

7. Let $\boldsymbol{A}$ be an orthogonal matrix of order 2.

(1) If $\det(\boldsymbol{A}) = 1$, then $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ for some θ.

(2) If $\det(\boldsymbol{A}) = -1$, then $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ for some θ.

Solution. Both (1) and (2) are true. Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 \end{pmatrix}$. Then $\|\boldsymbol{u}_1\| = 1$. Thus if θ is the angle between $\boldsymbol{u}_1$ and the positive x-axis, then $\boldsymbol{u}_1 = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$.

Let $\boldsymbol{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$. Then $0 = \boldsymbol{u}_1 \cdot \boldsymbol{u}_2 = x\cos\theta + y\sin\theta$ and $1 = \|\boldsymbol{u}_2\|^2 = x^2 + y^2$. Solving, one gets $y = \pm\cos\theta$ and $x = \mp\sin\theta$, and it's easy to verify that

$$\det(\boldsymbol{A}) = \begin{cases} 1, & \text{if } x = -\sin\theta, \quad y = \cos\theta; \\ -1, & \text{if } x = \cos\theta, \quad y = -\cos\theta. \end{cases}$$

8. Let $\boldsymbol{A}$ be an orthogonal matrix of order n.

(1) If λ is an eigenvalue of $\boldsymbol{A}$, then $\lambda = 1$ or $\lambda = -1$.

(2) 1 or -1 is an eigenvalue of $\boldsymbol{A}$.

Solution. (1) True. If $\boldsymbol{u}$ is an eigenvector of $\boldsymbol{A}$ with eigenvalue λ, then $\boldsymbol{Au} = \lambda\boldsymbol{u}$, and so $(\boldsymbol{Au})^{\mathrm{T}}\boldsymbol{Au} = (\lambda\boldsymbol{u})^{\mathrm{T}}(\lambda\boldsymbol{u}) = \lambda^2\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}$. But we also have $(\boldsymbol{Au})^{\mathrm{T}}\boldsymbol{Au} = \boldsymbol{u}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{Au} = \boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}$ since $\boldsymbol{A}$ is orthogonal. Thus $\lambda^2\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} = \boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}$, and so $\lambda^2 = 1$ since $\boldsymbol{u} \neq \boldsymbol{0}$. Hence $\lambda = \pm 1$.

(2) False. $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is orthogonal and has no (real) eigenvalues.

9. Let $\boldsymbol{A}$ be a square matrix of order n.

(1) If 0 is an eigenvalue of $\boldsymbol{A}$, then $\boldsymbol{A}$ is singular.

(2) If $\boldsymbol{A}$ is singular, then 0 is an eigenvalue of $\boldsymbol{A}$.

Solution. Both (1) and (2) are true; see Pg 11 of Chapter 6.

10. Let $\boldsymbol{A}$ be a square matrix of order n.

(1) If $\boldsymbol{A}$ is diagonalizable, then all the roots of the characteristic equation of $\boldsymbol{A}$ are real.

(2) If all the roots of the characteristic equation of $\boldsymbol{A}$ are real, then $\boldsymbol{A}$ is diagonalizable.

Solution. (1) True. If $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$ with

$$\boldsymbol{D} = \begin{pmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix},$$

then the characteristic polynomial of $\boldsymbol{A}$ is $(x - d_1)(x - d_2)\cdots(x - d_n)$.

(2) False. Let $\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then its characteristic polynomial is $x^2 = (x-0)^2$, but $\boldsymbol{A}$ cannot be diagonalized.